Dietary fiber, weight gain, and cardiovascular disease risk factors in young adults.

CONTEXT: Dietary composition may affect insulin secretion, and high insulin levels, in turn, may increase the risk for cardiovascular disease (CVD). OBJECTIVE: To examine the role of fiber consumption and its association with insulin levels, weight gain, and other CVD risk factors compared with other major dietary components. DESIGN AND SETTING: The Coronary Artery Risk Development in Young Adults (CARDIA) Study, a multicenter population-based cohort study of the change in CVD risk factors over 10 years (1985-1986 to 1995-1996) in Birmingham, Ala; Chicago, III; Minneapolis, Minn; and Oakland, Calif. PARTICIPANTS: A total of 2909 healthy black and white adults, 18 to 30 years of age at enrollment. MAIN OUTCOME MEASURES: Body weight, insulin levels, and other CVD risk factors at year 10, adjusted for baseline values. RESULTS: After adjustment for potential confounding factors, dietary fiber showed linear associations from lowest to highest quintiles of intake with the following: body weight (whites: 174.8-166.7 lb [78.3-75.0 kg], P<.001; blacks: 185.6-177.6 lb [83.5-79.9 kg], P = .001), waist-to-hip ratio (whites: 0.813-0.801, P = .004; blacks: 0.809-0.799, P = .05), fasting insulin adjusted for body mass index (whites: 77.8-72.2 pmol/L [11.2-10.4 microU/mL], P = .007; blacks: 92.4-82.6 pmol/L [13.3-11.9 microU/mL], P = .01) and 2-hour postglucose insulin adjusted for body mass index (whites: 261.1-234.7 pmol/L [37.6-33.8 microU/mL], P = .03; blacks: 370.2-259.7 pmol/L [53.3-37.4 microU/mL], P<.001). Fiber was also associated with blood pressure and levels of triglyceride, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, and fibrinogen; these associations were substantially attenuated by adjustment for fasting insulin level. In comparison with fiber, intake of fat, carbohydrate, and protein had inconsistent or weak associations with all CVD risk factors. CONCLUSIONS: Fiber consumption predicted insulin levels, weight gain, and other CVD risk factors more strongly than did total or saturated fat consumption. High-fiber diets may protect against obesity and CVD by lowering insulin levels.